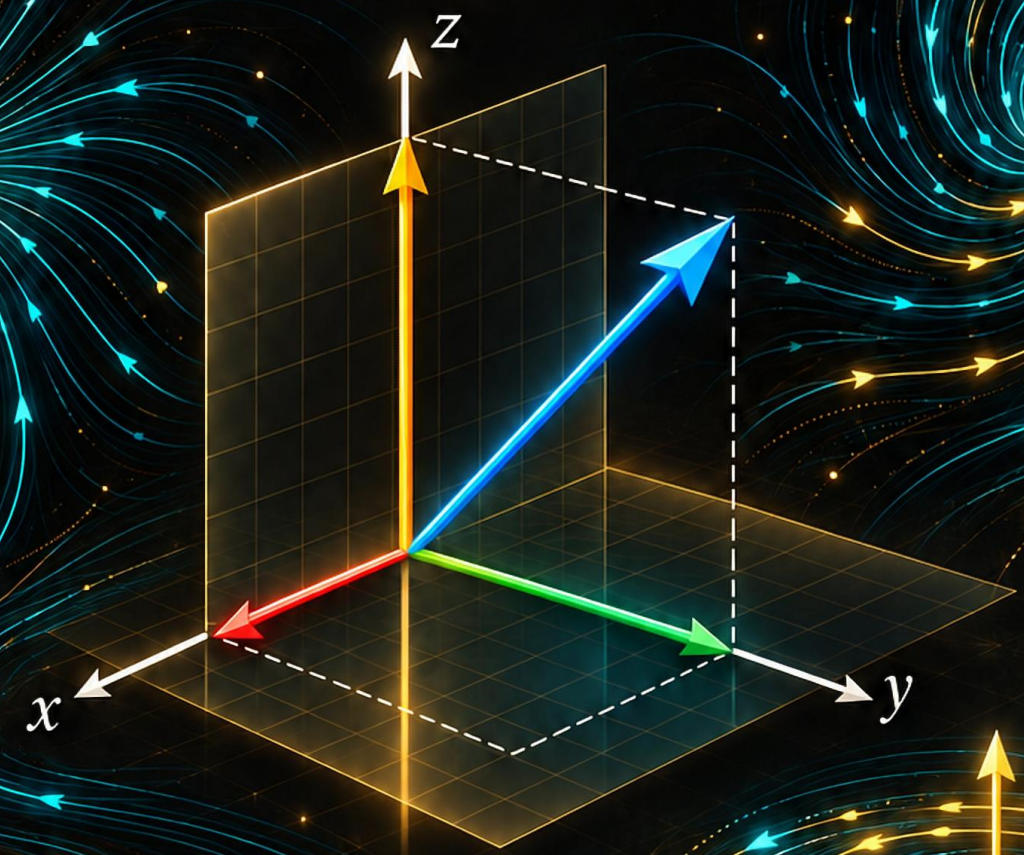
ELECTROMAGNETIC THEORY

CHAPTER 1

VECTOR ANALYSIS



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CHAPTER 1 · VECTOR ANALYSIS

Preface to Chapter 1

Electromagnetic theory describes the world in terms of *fields* — the electric field **E**, the magnetic field **H** and the potentials from which they are derived — and every one of its laws is written in the language of vectors. A student who is fluent in that language finds that Gauss's law, Ampère's law and, in the end, Maxwell's equations read almost like sentences; a student who is not spends much of the course wrestling with notation instead of physics. This is why the course opens with vector analysis.

The chapter is taught as eight lectures in two blocks. Lectures 1–4 develop *vector algebra*: scalars, vectors and fields; addition, subtraction and multiplication by a scalar; and the dot and cross products. Lectures 5–8 develop *vector calculus*: the del operator and the gradient, the divergence, the curl and the Laplacian. All the work is done in Cartesian coordinates, so that attention stays on the meaning of each operation rather than on the geometry of a particular coordinate system.

Line and surface integrals enter where the physics first needs them: the flux through a closed surface defines the divergence (Lecture 6) and the circulation around a closed path defines the curl (Lecture 7), each with a first look at the integral theorem it leads to — the divergence theorem and Stokes's theorem.

The lectures are written with the rest of the course in mind. The gradient returns as $\mathbf{E} = -\nabla V$ in Chapter 4; the divergence gives Gauss's law its point form, $\nabla \cdot \mathbf{D} = \rho_v$; the curl appears in Faraday's law and the Ampère–Maxwell law (Chapter 8) and in Ampère's law for steady currents (Chapter 10); and the Laplacian is the operator of Laplace's and Poisson's equations. Each such link is pointed out in a *Physical meaning* box. The notation — bold type for vectors and $\mathbf{a}_x$, $\mathbf{a}_y$, $\mathbf{a}_z$ for the Cartesian unit vectors — is that of Sadiku's *Elements of Electromagnetics*, one of the course textbooks.

To the student. Vector analysis rewards practice more than any other part of the course: the operations are few, but they must become automatic. Read the learning outcomes before each lecture, work every example yourself before reading its solution, and do the practice problems without looking at the answers. A problem you cannot start is a signal to return to the definitions and key results of that lecture.

Further reading for this chapter

- M. N. O. Sadiku, *Elements of Electromagnetics*, Oxford University Press — Chapters 1 and 3.
- W. H. Hayt and J. A. Buck, *Engineering Electromagnetics*, McGraw-Hill — Chapter 1 (vector algebra).
- D. J. Griffiths, *Introduction to Electrodynamics*, Cambridge University Press — Chapter 1.
- M. R. Spiegel, S. Lipschutz and D. Spellman, *Vector Analysis* (Schaum's Outline Series), McGraw-Hill — for further solved problems.

CHAPTER 1

Vector Analysis at a Glance

A weather map shows the temperature of every city as a single number, but the wind needs an arrow: how fast *and* which way. The same distinction runs through electromagnetism. The electric potential V is a number at every point; the electric field $\mathbf{E}$ and the magnetic field $\mathbf{H}$ are arrows. This chapter builds the tools for handling both — first the algebra of vectors, then their calculus — and every later chapter of the course is written with these tools.

Where this chapter leads

The operations of Lectures 5–8 are exactly what is needed to read Maxwell's equations in point form — the centre of the course, reached in Chapter 8:

Equation	Law	Built from
$\nabla \cdot \mathbf{D} = \rho_v$	Gauss's law	divergence, Lecture 6
$\nabla \cdot \mathbf{B} = 0$	No magnetic charge	divergence, Lecture 6
$\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$	Faraday's law	curl, Lecture 7
$\nabla \times \mathbf{H} = \mathbf{J} + \partial \mathbf{D} / \partial t$	Ampère–Maxwell law	curl, Lecture 7

In electrostatics the same tools give the field from the potential, $\mathbf{E} = -\nabla V$ (the gradient, Lecture 5), and Poisson's equation $\nabla^2 V = -\rho_v / \varepsilon$ (the Laplacian, Lecture 8).

Lectures in this chapter

Lecture	Topic	Central formula	Page
1	Scalars, vectors, fields and unit vectors	$\mathbf{a}_A = \mathbf{A}/A$	4
2	Addition, subtraction and scalar multiplication	$\mathbf{A} \pm \mathbf{B} = (A_x \pm B_x)\,\mathbf{a}_x + \cdots$	9
3	The dot (scalar) product	$\mathbf{A} \cdot \mathbf{B} = AB\cos\theta_{AB}$	13
4	The cross (vector) product	$\mathbf{A} \times \mathbf{B} = AB\sin\theta_{AB}\,\mathbf{a}_n$	17
5	The del operator and the gradient	∇V	23
6	The divergence of a vector field	$\nabla \cdot \mathbf{A}$	28
7	The curl of a vector field	$\nabla \times \mathbf{A}$	33
8	The Laplacian of a scalar field	$\nabla^2 V = \nabla \cdot \nabla V$	38

Reference pages: Formula Sheet, page 42; Answers to Practice Problems, page 43.

Before you begin

You will need partial differentiation, right-angle trigonometry and Pythagoras' theorem in three dimensions. Keep the right-handed x, y, z axes in mind: every result in the chapter is built on them.

Lecture 1 · Scalars, Vectors, Fields and Unit Vectors

Learning outcomes

By the end of this lecture you should be able to:

- distinguish scalar from vector quantities and give physical examples of each;
- explain what is meant by a scalar field and a vector field;
- write a vector in Cartesian component form and calculate its magnitude;
- construct the unit vector in the direction of any non-zero vector.

1.1 Why we need vectors

Electromagnetic theory is essentially the study of *fields*: the electric field intensity **E** and the magnetic field intensity **H** have a size *and* a direction at every point in space. To state the laws of electromagnetism compactly, and in a form that does not depend on how we happen to orient our axes, we need the language of vectors. Vector analysis is that language. This chapter develops its algebra (Lectures 1–4) and its calculus (Lectures 5–8).

1.2 Scalars and vectors

Definition — scalar. A *scalar* is a quantity that has only magnitude.

Time, mass, distance, temperature, entropy, electric potential and population are scalars. A scalar is fully specified by one number and a unit, for example $T = 25\ ^\circ\mathrm{C}$.

Definition — vector. A *vector* is a quantity that has both magnitude and direction.

Velocity, force, displacement and electric field intensity are vectors. To specify a vector we must give how large it is *and* which way it points.

A more general class of physical quantities, called *tensors*, includes scalars (tensors of rank 0) and vectors (tensors of rank 1) as special cases. In this course we work only with scalars and vectors.

Notation. A vector is written either with an arrow, $\vec{A}$, or in boldface, **A**. A scalar is written as a plain italic letter: A, B, U, V. In handwriting always use the arrow — a bold letter cannot be seen on paper, and dropping it turns a vector into a scalar.

Watch out. Having a "direction" is not enough to make something a vector: a vector must also *add* by the parallelogram rule of Lecture 2. The current I in a wire has a sense of direction, but currents meeting at a junction add as ordinary numbers, so I is a scalar. (The current *density* **J**, in $\mathrm{A/m^2}$, is a genuine vector.)

1.3 Scalar and vector fields

Definition — field. A *field* is a function that specifies a particular quantity everywhere in a region. If the quantity is a scalar, the field is a *scalar field*; if it is a vector, the field is a *vector field*.

Examples of scalar fields are the temperature distribution in a building, the sound intensity in a theatre, the electric potential in a region, and the refractive index of a stratified medium. Examples of vector fields are the gravitational force on a body in space and the velocity of raindrops in the atmosphere. Figure 1.1 shows the difference: a scalar field attaches one number to each point, a vector field attaches an arrow.

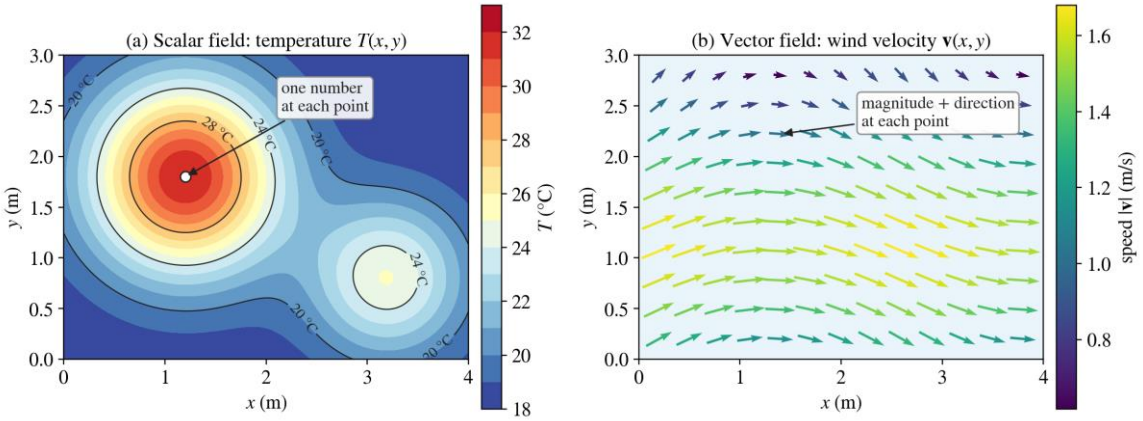


Figure 1.1 (a) A scalar field: the temperature $T(x, y)$ is a single number at every point (colour). (b) A vector field: the wind velocity $\mathbf{v}(x, y)$ has a magnitude (colour, arrow length) and a direction (arrow) at every point.

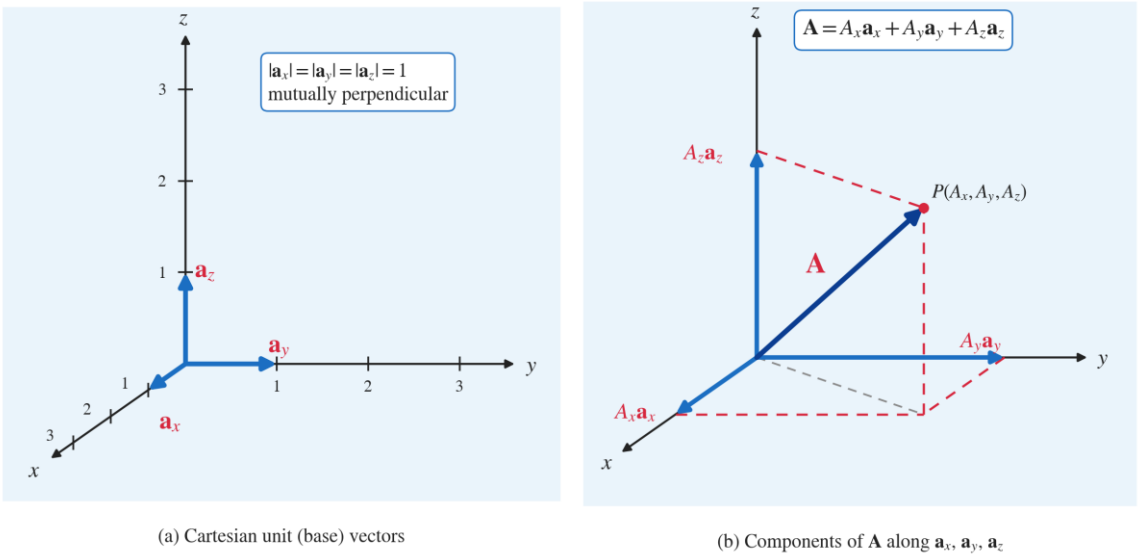
1.4 Cartesian unit vectors and components

The Cartesian (rectangular) coordinate system has three mutually perpendicular axes x, y, z arranged in a right-handed way. Along the three axes we place the base unit vectors $\mathbf{a}_x$, $\mathbf{a}_y$ and $\mathbf{a}_z$ (Figure 1.2a). Each has length 1 and points in the positive direction of its axis.

Any vector $\mathbf{A}$ can be built from three pieces, one along each axis (Figure 1.2b):

$$\mathbf{A} = A_x\,\mathbf{a}_x + A_y\,\mathbf{a}_y + A_z\,\mathbf{a}_z \qquad \text{or, briefly,} \qquad \mathbf{A} = \left(A_x,\ A_y,\ A_z\right)$$

The numbers A_x, A_y, A_z are the **components** of $\mathbf{A}$ in the x, y and z directions. They are scalars and may be positive, negative or zero. The vectors $A_x\mathbf{a}_x$, $A_y\mathbf{a}_y$, $A_z\mathbf{a}_z$ are the *vector components*.



(a) Cartesian unit (base) vectors

(b) Components of $\mathbf{A}$ along $\mathbf{a}_x$, $\mathbf{a}_y$, $\mathbf{a}_z$

Figure 1.2 (a) The Cartesian unit vectors $\mathbf{a}_x$, $\mathbf{a}_y$, $\mathbf{a}_z$. (b) The vector $\mathbf{A}$ is the sum of its three vector components $A_x\mathbf{a}_x$, $A_y\mathbf{a}_y$ and $A_z\mathbf{a}_z$; its tip is the point $P(A_x, A_y, A_z)$.

Magnitude. The length of $\mathbf{A}$ follows from applying Pythagoras' theorem twice in Figure 1.2b. The projection of $\mathbf{A}$ on the xy-plane (grey dashed line) has length $\sqrt{A_x^2 + A_y^2}$. This projection and the vertical piece A_z are perpendicular, so

$$A = |\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

The magnitude is never negative, and it is zero only for the zero vector $\mathbf{0} = (0,0,0)$.

1.5 The unit vector along a given vector

Definition — unit vector. A *unit vector* $\mathbf{a}_A$ along $\mathbf{A}$ is a vector whose magnitude is unity (i.e. 1) and whose direction is along $\mathbf{A}$.

Dividing a vector by its own length keeps the direction and makes the length 1:

$$\mathbf{a}_A = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\mathbf{A}}{A} = \frac{A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}, \qquad |\mathbf{a}_A| = 1$$

Rearranging gives the most useful way of thinking about any vector — *magnitude times direction*:

$$\mathbf{A} = A\,\mathbf{a}_A$$

This completely specifies $\mathbf{A}$ in terms of its magnitude A and its direction $\mathbf{a}_A$ (Figure 1.3).

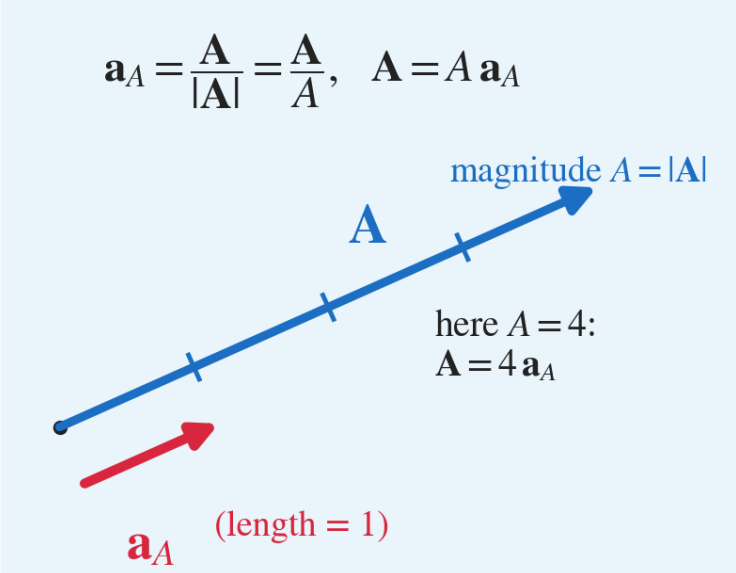


Figure 1.3 The unit vector $\mathbf{a}_A$ (red) has length 1 and the same direction as $\mathbf{A}$. Here $A = 4$, so $\mathbf{A}$ is four copies of $\mathbf{a}_A$ laid end to end.

Physical meaning. A unit vector is dimensionless: it carries direction only. All the physical units sit in the magnitude. For example, the force $\mathbf{F} = (20\ \text{N})\,\mathbf{a}_F$ says "20 newtons, in the direction $\mathbf{a}_F$". The zero vector has no direction, so it has no unit vector.

1.6 Worked examples

Worked Example 1.1 | Magnitude and unit vector

Find the magnitude of $\mathbf{A} = 3\mathbf{a}_x + 4\mathbf{a}_y - 12\mathbf{a}_z$ and the unit vector along it.

Solution. The magnitude is

$$A = \sqrt{3^2 + 4^2 + (-12)^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$$

so the unit vector is

$$\mathbf{a}_A = \frac{\mathbf{A}}{A} = \frac{3}{13}\mathbf{a}_x + \frac{4}{13}\mathbf{a}_y - \frac{12}{13}\mathbf{a}_z = 0.2308\,\mathbf{a}_x + 0.3077\,\mathbf{a}_y - 0.9231\,\mathbf{a}_z$$

Check: $|\mathbf{a}_A|^2 = (9 + 16 + 144)/169 = 1$ ✓.

Worked Example 1.2 | A vector of given length and direction

Find the vector of magnitude 15 that points in the direction of $\mathbf{B} = 2\mathbf{a}_x - \mathbf{a}_y + 2\mathbf{a}_z$.

Solution. First find the direction: $B = \sqrt{4 + 1 + 4} = 3$, so $\mathbf{a}_B = \frac{1}{3}(2\mathbf{a}_x - \mathbf{a}_y + 2\mathbf{a}_z)$. Then attach the required magnitude:

$$\mathbf{C} = 15\,\mathbf{a}_B = 10\,\mathbf{a}_x - 5\,\mathbf{a}_y + 10\,\mathbf{a}_z$$

Check: $|\mathbf{C}| = \sqrt{100 + 25 + 100} = 15$ ✓.

Worked Example 1.3 | Evaluating fields

The temperature in a room is $T(x,y,z) = 20 + 2x^2y$ (°C, with x, y, z in metres) and the air velocity is $\mathbf{v} = y\,\mathbf{a}_x - x\,\mathbf{a}_y$ (m/s). Evaluate both fields at the point (1,2,0) and state which is a scalar field and which is a vector field.

Solution. $T(1,2,0) = 20 + 2(1)^2(2) = 24$ °C — one number, so T is a scalar field. $\mathbf{v}(1,2,0) = 2\,\mathbf{a}_x - 1\,\mathbf{a}_y$ m/s — an arrow, so $\mathbf{v}$ is a vector field. Its speed there is $|\mathbf{v}| = \sqrt{4+1} = \sqrt{5} = 2.236$ m/s.

Summary of Lecture 1

- A scalar has magnitude only; a vector has magnitude and direction and adds by the parallelogram rule.
- A field assigns a scalar (scalar field) or a vector (vector field) to every point of a region.
- $\mathbf{A} = A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z$ and $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$.
- $\mathbf{a}_A = \mathbf{A}/A$ has unit length; $\mathbf{A} = A\,\mathbf{a}_A$ = magnitude × direction.

Practice problems

P1.1 Classify each quantity as a scalar or a vector: (a) kinetic energy, (b) momentum, (c) electric potential, (d) magnetic flux density, (e) temperature, (f) acceleration, (g) electric charge, (h) the current in a wire.

P1.2 For $\mathbf{A} = 2\mathbf{a}_x - 3\mathbf{a}_y + 6\mathbf{a}_z$, find $|\mathbf{A}|$ and $\mathbf{a}_A$.

P1.3 Find the vector of magnitude 20 in the direction of $\mathbf{B} = \mathbf{a}_x - 2\mathbf{a}_y + 2\mathbf{a}_z$.

P1.4 Show that $\mathbf{u} = \cos\alpha\,\mathbf{a}_x + \sin\alpha\cos\beta\,\mathbf{a}_y + \sin\alpha\sin\beta\,\mathbf{a}_z$ is a unit vector for all angles α and β.

P1.5 A plate has temperature $T = 30 - x^2 - 2y^2$ (°C). (a) Evaluate T at (1,1,0) and at (2,0,5). (b) A fluid on the plate moves with $\mathbf{v} = x\,\mathbf{a}_x + 2y\,\mathbf{a}_y$ m/s. Find the speed at (3,2,0).